

MADE IN INDIA



UNIK SURGICAL PVT. LTD.

Manufacturer, Importer, Exporter & Suppliers

Medical Furniture, Lab Furniture, Non Medical
Furniture, MOT, MGPS, Laundry Equipments, Kitchen Equipments, CSSD, TSSU,
BMW, Medical Equipments, Mortuary Equipments, Nurse Call System,
Lab Equipments & General Orders Supplier



CSSD

CENTRAL STERILE
SERVICES DEPARTMENT



QUALITY | TECHNOLOGY
COMMITMENT | SERVICES



UNIK SURGICAL PVT. LTD.

Founded in 1999, Unik Surgical Pvt. Ltd. has been at the forefront of healthcare infrastructure development in Eastern India. What began as a small enterprise has grown into a leading supplier and manufacturer of medical and surgical equipment and furniture.

With our headquarters in Patna, Bihar, a regional office in Greater Noida, Uttar Pradesh, and a manufacturing facility in Chapra, Bihar, we serve clients across the region with unmatched expertise and a strong commitment to quality. Our journey is defined by continuous innovation, strict adherence to international standards, and an unwavering dedication to improving healthcare facilities across India.

Contracting

Project Report

Layout

Configuration

Consulting

Installation

On-site Training

Startup

Costing

Technical Support

ABOUT DIRECTOR

Manoj Kumar Singh is a healthcare industry leader known for his practical approach and deep commitment to improving patient care. With years of hands-on experience in medical equipment manufacturing, he founded Unik Surgical with a simple goal — to make reliable, high-quality surgical solutions accessible to healthcare facilities of all sizes.

What began as a trader in Patna has now grown into a trusted name, serving hospitals and clinics across the country. Manoj's focus on precision engineering, transparent business ethics, and long-term client relationships has helped Unik Surgical earn the confidence of medical professionals.

His leadership continues to guide the company toward innovation and responsible growth, while staying rooted in the belief that every piece of equipment should ultimately contribute to better patient outcomes.



International & National Accreditation of Bodies



एन. एस. आई. सी.
NSIC
ISO 9001 : 2008

CERTIFICATIONS



QUALITY | TECHNOLOGY
COMMITMENT | SERVICES

MODEL NO : USPL-USS-105

USPL H.P. CYLINDRICAL HORIZONTAL STERILIZER

Introduction :

The sterilizer is based on the principal that microbiological organism are killed or made inactive by maintaining high temperature for related time. The medium use for raising the temperature is dry saturated steam. The steam also displace the air under the gravity effect out of the chamber resulting efficient sterilization.

Robust and rigid construction, designed for all types of bulk sterilization which are commonly needed in Medical, Agricultural and Pharmaceutical Institutions.

Suitable for Sterilizing hospital dressings, linen, rubber, plastic goods, surgical instruments, glass wares, utensils etc. Construction : Triple walled with steam jacket and separate boiler Inner chamber and steam jacket are made up of Heavy gauge S.S. Sheet with leak proof argon-arc welding.

The sterilizer has single piece door made of steel plate lined with S.S. from inside and back plate and ring is also made of thick stainless steel sheet. All the sterilizers are hydraulically tested to withstand 2.5 times the working pressure. Door is grooved to hold EPDM or silicon (Optional) gasket to securely rest on front facia of door ring for easy & perfect leak proof locking.

Mounted on tubular steel frame with ground leveling screwed flanges. The Outer jacket is wrapped with insulation material to minimise the heat losses due to radiation and is covered by polished stainless steel sheet for elegant appearance.

Steam Generator : Made of heavy stainless steel sheet and provided in unique design with minimum joints and bolted parts thus reduces the chances of leaks. Suitable capacity heating elements and low water level cutoff system to protect the former from burning out dry are provided. Main hole provided on heater plate provides for easy cleaning of the deposited scale inside the generators for better efficiency. Fitted with gauge glass assembly with shut off valves (to avoid spillage of hot water in case of glass tube breakage) for water level indication, water inlet and outlet valves.

Pressure Control : The "PIEZOSTAT" (Pressure Control Device) is incorporated in all electrically operated sterilizers. It economizes on power consumption and control the steam generation to set value depending on the temperature for sterilization

Safety Door : Sterilizer has hinged self locking single piece thick steel plate lined with S.S. from inside door and cannot be opened when under pressure. The self locking device automatically engages the clutch mechanism when the chamber is pressurized to avoid opening of the door thus eliminates the chances of any accident.

Single Point Control : Complete sterilizer cycle is controlled from one



point with the help of " Multiport valve" fitted at the top. Two speeds of steam exhaust are available, fast and slow. Drying of the sterilized load can also be done.

Easily readable jacket and chamber pressure gauges are mounted along the multi port valve. A steam trap is fitted in the chamber discharge line to drain the condensate automatically. Switch Box: Houses the PIEZOSTAT, main switch, pilot lamps, relays etc. and mounted on the stand of the sterilizer for easy operation.

Other Features:

- **TRIPLE SAFETY:** The Sterilizer is provided with the triple safety features. At the boiler level by a PIEZOSTAT which automatically limits the pressure to the required set value and a spring loaded safety valve in case of its failure which releases the steam to keep pressure within the safety limits. At the chamber level, a spring loaded safety valve and a dead weight release valve to release the steam in case the pressure exceeds the safety limit. All these safety features function independent of one another and subsequently i.e. one takes over the charge in case of other's failure.
- A plug screen prevents the discharge line from choking. The plug is easily removable for daily cleaning.
- Powerful Ejector for drying sterilized linen circulates air throughout the chamber. The circulating air passes through a corrosion resistant metallic wool filter.
- A dial thermometer is provided to show the chamber temperature.
- Automatic Vacuum breaker is provided to break vacuum in case of formation of vacuum due to steam condensation.

Operating Temp. & Pressure :
 Sterilizing Temp : 121°C
 Sterilizing pressure : 1.2 to 1.5 kg/cm² (15 psi to 22psi)
 Power Requirement : Suitable to operate on 440Vvolts, 3 ph, 50 Hz, Ac supply

Available Models :

Standard : The autoclave is manually operated

Semi Automatic : The process is controlled by programmable logic controller & is user friendly. Four fixed sterilization cycles take care of hospital sterilization needs. The automation avoids the human errors and interference thus assures quality sterilization. This unit can be handled by unskilled person and need not to be attended all the time thus sparing the operator for other work.

Fully Automatic : This is the upper version of semi automatic machine with five fixed and one variable parameters cycles. Main machine interface is provided to monitor the process parameters, set the desired parameters and to get online printing of the process to keep records for future reference.

Material of Construction :

	Std.	Optional
Chamber	-S.S.304	S.S.316/S.S.316L/S.S.316Ti
Jacket	-S.S.304	S.S.316/BQ MS
Door	-BQ MS	S.S.304/S.S.316
Steam Generator	-S.S.304	S.S.316
Outer Cover	-S.S.304	-----
Stand	-ERW M	S S.S.304

Tubes

Optional Accessories :

- Digital Temperature Indicator can be supplied in lieu of dial thermometer.
- Digital Temperature Controller with timer can be fixed at an extra cost. Once the temperature is attained it will automatically start the timer. It raises an alarm at a set time, so that operator is alerted & he can perform balance operations
- Automatic water feed system can be incorporated in the boiler, so that at higher level it switched off the water pump and restart it at lower level automatically
- In addition to vacuum dry provided in the MPV an extra ejector can be provided in the autoclave for better drying of the load.
- Digital Pressure & Compound gauges can be provided in lieu of analog gauges.
- POP type safety valve can be installed in lieu of spring load safety valve, for more accurate & better performance if required, at an extra cost.

- Dressing Drums
- Parisian for Direct steam supply from centralized boiler
- Temperature recorder
- High pressure High vacuum model
- Additional door (Double door) model.
- Fully automatic micro processor controlled model

Spare Accessories :

- Pressure gauge/Compound gauge
- Water level gauge glass
- Dial thermometer
- Non return valve
- Moisture trap
- Jointless Gasket : Neoprene/silicon rubber,
Size : 400mm & 500mm
- Heating elements (Flange Type)
Rating kw 6.0, 9.0

Various level of automation is available to suit customer need.

- | | |
|-------------------|--|
| (1) USPL-USS-5.0 | Four process cycles provided are with fixed parameters and takes care of all sterilization need. |
| - | No display is available. |
| - | No online data printing |
| (2) USPL-USS-7.5 | Four process cycles with fixed parameter. |
| - | No display |
| - | Online process printing is available. |
| (3) USPL-USS-10.0 | Four process cycles with fixed parameters. |
| - | Display (Two line) |
| - | Online process printing |
| (4) USPL-USS-15.0 | Five fixed and one flexible process cycles are provided. |
| - | Display (Four line) |
| - | Online printing |
| (5) USPL-USS-20.0 | Five fixed and one flexible process cycle. |
| - | Display (four line/ Touch Screen) |
| - | Online printing |
| - | High end PLC is provided to take |

Available in following capacities :

dia	x	depth	load
400	x	600 mm	6 kw
400	x	1100 mm	9kw
500	x	900 mm	9kw
500	x	1200 mm	9kw
600	x	900 mm	9kw
750	x	1100mm	18 kw

MODEL NO. USPL-USS-106

USPL ETHYLENE OXIDE STERILIZER

Ethylene Oxide Sterilizer suitable for sterilization of Catheters Disposable. The unit is made out of stainless steel 304 chambers. Stainless steel back plate as well as S.S./M.S. door. The unit is having individual valves for exhaust, allowing gas in the chamber and for vacuumising. The unit is having vacuum pump, chamber heating and control system. Table top model suitable for 2 Nos. Sterilizing Baskets. The unit is having fascia panel. All accessories pump etc. are enclosed with stainless steel cover. Gauges and controls are provided in the fascia panel. This model is specially designed for Hospitals and Research Lab. The unit is suitable for 10 : 90 EO : CO_2 , or 12 : 88 EO : F-12 mixture.

Table top model are having following sizes :

- Cylindrical Model
300 mm (Dia) x 500 mm (Depth)
450 mm (Dia) x 450 mm (Depth)
- Rectangular Model
250 (H) x 300 (W) x 550 (D) mm.
250 (H) x 300 (W) x 800 (D) mm.



	Ethylene oxide	Diluent gas	Flammability
EO	100%	Nil	Flammable
12EO:88F	12%	88% Freon	Non Flammable
10EO:90C	10%	90% CO_2	Non Flammable
20EO:80C	20%	80% CO_2	Non Flammable
80EO:20C	80%	20% CO_2	Flammable
90EO:10C	90%	10% CO_2	Flammable

These highly sophisticated and rugged can be suitably use for

- Medical Industry
- Pharmaceutical Industry
- Spices & Herbal Industry
- Cosmetic Industry
- Electronics Industry

Complete range of Product & Services.

- Sterilizers
- Accessories & Ancillary Equipment
- EO Gas Mixture Cylinders / Disposables Cartridge
- Design of Exhaust & Aeration System
- Process Validation as ISO / CE Norms
- Calibrations

MODEL NO. USPL-USS-106

USPL ETHYLENE OXIDE STERILIZER

Free standing bulk model : Suitable for large Hospitals & for small manufacturing units. Suitable for 10/90 EO/CO₂ mixture or 90/10 EO/CO₂.

The System can be provided with manual and automatic Systems with single or double door combination. Low cost automation along with data logging facility can also be incorporated to maintain process stability and hard copy evidence.

Table top model are having following sizes :

- Cylindrical Model
400 mm (Dia) x 600 mm (Depth)
500 mm (Dia) x 1200 mm (Depth)
- Rectangular Model
600 (H) x 600 (W) x 1200 (D) mm.
900 (H) x 600 (W) x 1500 (D) mm.
Other sizes on request.

These highly sophisticated and rugged can be suitably use for

- Medical Industry
- Pharmaceutical Industry
- Spices & Herbal Industry
- Cosmetic Industry
- Electronics Industry

Complete range of Product & Services.

- Sterilizers
- Accessories & Ancillary Equipment
- EO Gas Mixture Cylinders / Disposables Cartridge
- Design of Exhaust & Aeration System
- Process Validation as ISO / CE Norms
- Calibrations



	Ethylene oxide	Diluent gas	Flammability
EO	100%	Nil	Flammable
12EO:88F	12%	88% Freon	Non Flammable
10EO:90C	10%	90% CO ₂	Non Flammable
20EO:80C	20%	80% CO ₂	Non Flammable
80EO:20C	80%	20% CO ₂	Flammable
90EO:10C	90%	10% Co ₂	Flammable

MODEL NO. USPL-USS-107

USPL H.P. HIGH SPEED CYLINDRICAL HORIZONTAL STERILIZER

As per IS-3829 II

Introduction :

The sterilizer is based on the principal that microbiological organism are killed or made inactive by maintaining high temperature for related time. The medium use for raising the temperature is dry saturated steam. The steam also displace the air under the gravity effect out of the chamber resulting efficient sterilization.

Robust and rigid construction, designed for all types of bulk sterilization which are commonly needed in Medical, Agricultural and Pharmaceutical Institutions.

Suitable for Sterilizing hospital dressings, linen, rubber, plastic goods, surgical instruments, glasswares, utensils etc.

Construction :

Triple walled with steam jacket and separate boiler Inner chamber and steam jacket are made up of Heavy gauge S.S. Sheet with leak proof argon-arc welding.

The sterilizer has single piece door made of steel plate lined with S.S. from inside and back plate and ring is also made of thick stainless steel sheet. All the sterilizers are hydraulically tested to withstand 2.5 times the working pressure. Door is grooved to hold EPDM or silicon (Optional) gasket to securely rest on front facia of door ring for easy & perfect leak proof locking.

Mounted on tubular steel frame with ground levelling screwed flanges. The Outer jacket is wrapped with insulatoin material to minimise the heat losses due to radiation and is covered by polished stainless steel sheet for elegant appearance.

Steam Generator : Made of heavy stainless steel sheet and comes in unique design with minimum joints and bolted parts thus reduces the chances of leaks suitable capacity heating elements and low water level cutoff system of protect the former from burning out dry are provided. Main hole provided on heater plate provides for easy cleaning of the deposited scale inside the generators for better efficiency. Fitted with gauge glass assembly with shut off valves (to avoid spillage of hot water in case of glass tube breakage) for water level indication. Water inlet and outlet valves.



Pressure Control : The " PIEZOSTAI" (Pressure Control Device) is incorporated in all electrically operated sterilizers. It economizes on power consumption and control the steam generation to set value depending on the temperature for sterilization.

Safety Door : Sterilizer has hinged self locking single piece thick steel plate lined with S.S. from inside door and cannot be opened when under pressure. The self locking device automatically engages the clutch mechanism when the chamber is pressurized to avoid opening of the door thus eliminates the chances of any accident.

Single Point Control: Complete sterilizer cycle is controlled from one point with the help of " Multi port valve" fitted at the top. Two speeds of steam exhaust are available, fast and slow. Drying of the sterilized load can also be done.

Easily readable jacket and chamber pressure gauges are mounted along the multi port valve. A steam trap is fitted in the chamber discharge line to drain the condensate automatically.

Switch Box:

Houses the PIEZOSTAT, main switch, pilot lamps, relays etc. and mounted on the stand of the sterilizer for easy operation.

Other Features :

- **TRIPLE SAFETY :** The Sterilizer is provided with the triple safety features. At the boiler level by a PIEZOSTAT which automatically limits the pressure to the required set value and a spring loaded safety valve in case of its failure which releases the steam to keep pressure within the safety limits. At the chamber level, a spring loaded safety valve and a dead weight release valve to release the steam in case the pressure exceeds the safety limit. All these safety features function independent of one another and subsequently i.e. one takes over the charge in case of other's failure.
- A plug screen prevents the discharge line from choking. The plug is easily removable for daily cleaning.
- Powerful Ejector for drying sterilized linen circulates air throughout the chamber. The circulating air passes through a corrosion resistant metallic wool filter.
- A dial thermometer is provided to show the chamber temperature.
- Automatic Vacuum breaker is provided to break vacuum in case of formation of vacuum due to steam condensation.

Operating Temperature & Pressure :

Sterilization Temperature : 134°C - 138°C

Sterilization Pressure : 1.2 to 2.4 Kg/cm² (30 PSI to 35 PSI)

Power Requirement : Suitable to operate on 440 V, 3 ph, 50 Hz. AC Supply.

Available Models :

Standard : The autoclave is manually operated. All the functions are carried out with the help of multiport valve.

Semi Automatic : The process is controlled by programmable logic controller & is user friendly. Four fixed sterilization cycles take care of hospital sterilization needs. The automation avoids the human errors and interference thus assures quality sterilization.

This unit can be handled by unskilled person and need not to be attended all the time thus sparing the operator for other work.

Fully Automatic : This is the upper version of semi automatic machine with five fixed and one variable parameter cycles. Main machine interface is provided to monitor the process parameters, set the desired parameters and to get online printing of the process to keep records for future reference.

Capacity:	Dia	x	Depth	Load
	400	x	600 mm	18 kw
	500	x	900 mm	36 kw

Material of Construction :

	Std.	Optional
Chamber	-S.S.304	S.S.316/S.S.316L/S.S.316Ti
Jacket	-S.S.304	S.S.316/BQ MS
Door	-BQ MS	S.S.304/S.S.316
Steam Generator	-S.S.304	S.S.316
Outer Cover	-S.S.304	-----
Stand	-ERW MS	S.S.304
Tubes		

Optional Accessories :

- Digital Temperature Indicator can be supplied in lieu of dial thermometer.
- Digital Temperature Controller with timer can be fixed at an extra cost. Once the temperature is attained it will automatically start the timer. It raises an alarm at a set time, so that operator is alerted & he can perform balance operations.
- Automatic water feed system can be incorporated in the boiler, so that at higher level it switched off the water pump and restart it at lower level automatically.
- In addition to vacuum dry provided in the MPV an extra ejector can be provided in the autoclave for better drying of the load.
- Digital Pressure & Compound gauges can be provided in lieu of analog gauges.
- Pop type safety valve can be installed in lieu of spring load safety valve, for more accurate & better performance if required, at an extra cost.
- Dressing Drums
- Provision for Direct steam supply from centralized boiler
- Temperature recorder
- High pressure High vacuum model
- Additional door (Double door) model.
- Fully automatic micro processor controlled model

Spare Accessories :

- Pressure gauge/Compound gauge
- Water level gauge glass
- Dial thermometer
- Non return valve
- Moisture trap
- Jointless Gasket :- Neoprene/silicon rubber, size :- 400mm & 500mm
- Heating elements (Flange Type) Rating kw 6.0, 9.0

Enquire for latest micro processor based fully automatic sterilizers. We also undertake Equipping of C.S.S.D. with equipments like Glove Washers, Dryers, Powders, Gauge Cutting Machine, Ultrasonic Cleaners, Scrub Stations, Sorting Tables, C.S.S.D. Racks, Baskets, Basket Racks and Distribution Trolleys.

MODEL NO. USPL-USS-108

USPL H.P. RECTANGULAR HORIZONTAL STERILIZER

Introduction : The sterilizer is based on the principle that microbiological organism are killed or made inactive by maintaining high temperature for related time. The medium used for raising the temperature is dry saturated steam. The steam also displaces the air under the gravity effect out of the chamber resulting efficient sterilization and is most economical way of bulk sterilization as required by medical pharmaceutical and agricultural institutes. These sterilizers are available as high pressure sterilizers and High Pressure High Vacuum sterilizers. In HPHV sterilizers forced vacuum is created to remove the trapped air from packs so that steam penetration is effective resulting proper sterilization. Suitable for sterilization of hospital dressing, linen, rubber plastic goods, surgical instruments, glassware and utensils etc.

Construction : The sterilizer's design is robust, energy efficient with high stability. The even steam circulation all around the material placed inside the chamber and even heating (no air pockets) is the overruling point of this design. Inner chamber is made out of thick plate of non magnetic stainless steel of grade SS 304/SS 316 fully. TIG welded. Jacket is of BQ steel as per IS 2002 /SS 304/ SS 316 Channel jacket usually preferred which is designed to give high pressure strength as well as to save energy.

The sterilizer will have single / (optional) double door made out of BQ steel lined with S.S. from inside or complete S.S. 304/316 (Optional). The door will be radial arm hinged door or (Optional) pneumatically operated vertically /horizontally sliding door. The shell is insulated with mineral/glass wool covered with CRCA sheet duly painted/ SS 304/ Aluminium. All the sterilizers are hydraulically tested to withstand 2.5 times the working pressure.

Steam Generator : Made of heavy stainless steel sheet and comes in unique design with minimum joints and bolted parts thus no chances of leaks. Suitable capacity heating elements and low water level cutoff system to protect the former from burning out dry are provided. Main hole provided on heater plate provides for easy cleaning of the deposited scale inside the generator for better efficiency. Fitted with gauge glass assembly with setoff valves (to avoid spillage of hot water in case of glass tube breakage) for water level indication, water



inlet and outlet valves.

Pressure Control : The "PIEZOSTAT" (pressure control device) is incorporated in all electrically operated sterilizers. It economizes on power consumption and control the steam generation to set value depending on the temperature for sterilization.

Safety Door : Doors have self pressure locking safety device and cannot be opened when under pressure. The self locking device automatically engages the clutch mechanism when the chamber is pressurized to avoid opening of the door thus eliminates the chances of any accident. The sliding door has (optional) inbuilt retraction mechanism which avoids any accident caused due to obstruction during its movement.

The doors are sealed with chamber by heat resistant EPDM / Silicon / Neoprene rubber gaskets.

Single Point Control : Complete sterilizer cycle is controlled from one point with the help of Multi port Valve "fitted at the top. Two speeds of steam exhaust are available fast and slow. Drying of the sterilized load can also be done. An additional ejector is also provided for better drying of the sterilized load. Easily readable jacket and chamber pressure gauges are mounted along the multi port valve. A steam trap is /fitted in the chamber discharge line to drain the condensate automatically.

Switch Box : Houses the Pressure control switch, Main switch, Pilot lamps, relays etc. and mounted on the stand of the sterilizer for easy operation.

Other Features :

- Triple Safety : The sterilizer is provided with triple safety.
 - (a) By a pressure control switch which automatically limits the pressure to the required set valve.
 - (b) Two spring loaded safety valves one each at chamber and boiler level.
 - (c) A dead weight safety valve at chamber level.
- A Plug Screen: Prevents the discharge line from choking. The plug is easily removable for routing cleaning.
- A powerful Ejector (for drying sterilized linen) circulates air through the chamber. The circulating air passes through a corrosion resistant metal wool filter.
- A Dial thermometer is provided to show the chamber temperature.
- Automatic Vacuum Breaker is provided to break vacuum in case of vacuum formation due to steam condensation in the jacket.

Operating Temperature and pressure :

High Pressure Sterilizer

- Sterilizing Temp.: 121°C
- Sterilizing Pressure : 1.2 to 1.5 kg/Cm² (15 psi to 22 psi)

Material of Construction :

	Std.	Optional
Chamber	-S.S.304	S.S.316/S.S.316L/S.S.316Ti
Jacket	-BQ MS	S.S.304/S.S.316
Door	-BQ MS	S.S.304/S.S.316
Steam Generator	-S.S.304	S.S.316
Outer Cover	-CRCA MS	S.S.304/Aluminium
Stand	-ERW MS	S.S.304
	Tubes/MS Channel	

Available in following capacities :

W	H	D	Load
450	450	900	18Kw
600	600	1200	18Kw
600	900	1500	36Kw
900	900	1500	36Kw
900	1050	2100	72Kw with auto water feeding (or direct steam operated)
1050	120	2100	72Kw with auto water feeding (or direct steam operated)

Various level of automation is available to suit customer need.

(1)

-No display is available.

-No online data printing

(2) -Four process cycles with fixed parameter.

-No display

-Online process printing is available.

(3) -Four process cycles with fixed parameters.

-Display (Two line)

-Online process printing

(4) - Five fixed and one flexible process cycles are provided.

-Display (Four line)

-Online printing

(5) - Five fixed and one flexible process cycle.

-Display (four line/ Touch Screen)

-Online printing

-High end plc is provided to take care of process parameter like Fo Value etc.

Option a /Accessories :

- Chamber/Jacket of S.S. 316
- Door of S.S. 304/316
- Additional Tray/Shelves
- Carriage & Trolley
- Parisian for direct steam supply from centralised boiler
- Temperature recorder
- High pressure High vacuum model
- Additional door (Double door) model.
- Fully automatic micro processor controlled model

Spare Accessories :

- Pressure gauge/Compound gauge
- Water level gauge glass
- Dial thermometer
- Non return valve
- Steam trap
- Jointless Gasket : Neoprene/Silicon rubber
- Heating elements (Flange Type) Rating kw 6.0, 9.0

**All models can have Box Type (European Style) design for elegant look (Optional)

**All other higher sizes as per requirement of the customer and will be direct steam (from centralized sources) operated.

USPL HIGH PRESSURE HIGH VACCUM HORIZONTAL RECTANGULAR STEAM STERILIZER

Construction :

- The chamber is sq. / rectangular in shape constructed of heavy duty stainless steel 304 / 316, door is also made of stainless steel 316L/316. Steam Channel jacket is made of Boiler grade steel sheet IS 2001or S.S. 304 Grade (Optional) Proper wall re-inforcement is provided on the chamber.
- The pipes and fittings are of stainless steel and bronze, the valves are ball type. Each sterilizer is mounted on a sturdy tubular type stand made of M.S. duly powder coated or (Optional) S.S.304 or modern type cubical panel (Optional) with easy access to servicing.
- All other construction details are as per IS 3829 Part-11999. Design :
- YSI Autoclave YSI-408 (HPHV) are designed as per International Standard
- Chamber of the autoclave is designed & tested as per ASME Std of hydraulic test pressure, two times the working pressure & jacket for 2 1/2 times or the working pressure performance as per BS & IS Standards
- Working pressure of autoclave is from 1.2 to 2.1 kg/cm² corresponding of the temperature of 121°C to 134°C and vacuum upto 26" HG (with built in vacuum pump arrangement) with necessary condenser etc.

Pulsating Deep Vacuum :

- High efficiency vacuum pump for mechanical air removal for excellent steam penetration even in fabric loads.
- Post sterilization vacuum drying treatment to ensure the load is dry on unloading.
- The chamber is designed and tested as per American Society of Mechanical Engineers standards.

Doors :

- Available Single and double door model offers the customer two door options - hinged door and automatic sliding door (Optional). All doors have stainless steel contact parts and temperature resistant silicon rubber gasket.

Recording System (Optional)

- Single point strip chart recorder for recording the process temperature, date and time.



Safety :

- Door safety to prevent starting of the process unless the door is closed and opening of the door when the chamber is pressurized (optional).
- Insulated chamber to avoid scalding to the operator.
- Safety valves for over-pressure safety.
- Pressure vessels hydro-tested for maximum quality assurance.
- Provided with two locks one automatic and another is manual. The automatic lock operates when pressure gauge reads 0.30-0.37kg/cm² in the chamber

Source of Steam Generation :

Following combination of heating models are available.

- Electrically operated
- Steam operated
- Combination of electric & steam operation

Steam Generator:

A stainless steel electric boiler is provided under the shell with interconnecting steam pipes. The boiler is supplied with 6 KW immersion heaters mounted on stainless steel plate and electric controls would comprise of an air break contactor, automatic pressure controller, low water protection. The boiler is also additionally provided with one independent spring - loaded safety valve, pressure gauge, gauge glass assembly with protector, water drain and water inlet valves.

Other Features :

- Two Nos. Mechanical ejectors
- Self Sterilizing vacuum drier
- Safety valve spring loaded and vacuum breaker
- Screen plug and Dial Thermometer for chamber's discharge line which is provided with steam trap and swing check valve
- Manual single point operation to operate manually as per IS 3829 Part-I
- Door safety -when the chamber is under pressure self locking device automatically locks and unlocks when chamber pressure is falls below 0.35 k.g./cm².
- Built in steam generator made of S.S. 304 with Low water cut out water level indicator self locking gauge glass valve.
- The equipment is provided with S.S. 304 railing and carriage with MS Trolley (Optional).
- Provided with vacuum system, low water cut off system.
- Suitable to run on in built electric Boiler as well as from centralized steam supply available near to equipment @ 2.1 to 2.5 kg/cm²
- Working temperature 134°C.

Utilities & Environment Conditions :

Supply Voltage 380/415 V3 Phase 50 Hz.

(for power refer chart)

- Feed Water : Soft water/ DM Water.

Sizes Available :

Width (mm)	Length (mm)	Height (mm)	Electric Load (K.W.)
500	500	950	18Kw
600	600	1200	18Kw
600	900	1200	36Kw
910	910	1200	36Kw
600	900	1500	36Kw

Other sizes as per customer requirement.

Various level of automation is available to suit customer need.

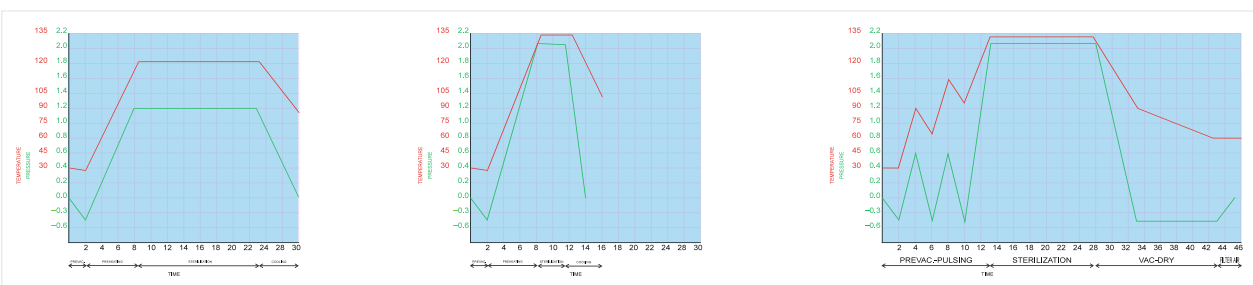
- (1) - Four process cycles provided are with fixed

parameter and takes care of all sterilization need.

- No display is available.
- No online data printing
- (2) - Four process cycles with fixed parameter.
- No display
- Online process printing is available.
- (3) - Four process cycles with fixed parameters.
- Display (Two line)
- Online process printing
- (4) - Five fixed and one flexible process cycles are provided.
- Display (Four line)
- Online printing
- (5) - Five fixed and one flexible process cycle.
- Display (four line/Touch Screen) Online printing
- High end plc is provided to take care of process parameter like Fo Value etc.

Other Optional Accessories :

- Sterilization Tray.
- Sterilization Basket.
- Carriage.
- Transfer Trolley.
- Air Compressor
- Audio visual alarm for low water level.
- Auto feed water system.
- Temperature recorder.
- Thermograph or Printer for documentation
- Others as per requirement of the customer.



MODEL NO. USPL-USS-205

USPL H.P. CYLINDRICAL HORIZONTAL STERILIZER

Construction:

- The Chamber is cylindrical in shape constructed out of heavy duty stainless steel 304/316 door and jacket of stainless steel 304 (316L or 316Ti-optional)
- The Pipes and fittings of stainless steel and bronze, the valves are ball type. Each sterilizer is mounted on a M.S/S.S sturdy stand. The outside panelling is made of M.S Powder coated or S.S to give elegant look. Hinged panels on sides makes it friendly for service purpose. All other construction details are as per IS 3892 Part 1.

Design :

- YSI autoclave (YSU-605) are designed as per international standard.
- Chamber of the autoclave is tested for hydraulic test pressure, performance as per BS & IS standards.
- Working pressure of autoclave is from 1.2 to 2.1 kg. cm² and vacuum upto 26"HG (with built in vacuum pump arrangement).
- Door gasket is expansion type silicon and jointless.
- The heating system provided is energy efficient and reduces running cost to great extent.



- Sterilization process is controlled by programmable logic controller with high accuracy & takes care of all user requirements.

Pulsating Deep Vacuum :

- High efficiency vacuum pump for mechanical air removal for excellent steam penetration even in fabric loads. Post sterilization vacuum drying treatment to ensure the load is dry on unloading.
- The chamber is designed and tested as per American Society of Mechanical Engineers standards.

Doors :

- Available Single and double door model offers the customer two door options - hinged door and automatic sliding door (Optional). All doors have stainless steel contact parts and temperature resistant silicon rubber gasket.

Recording System (Optional):

- Single point strip chart recorder for recording the process temperature, date and time.

Safety :

- Door safety to prevent starting of the process unless the door is closed and opening of the door when the chamber is pressurized (optional).
- Insulated chamber to avoid scalding to the operator.
- Safety valves for over-pressure safety.

- Pressure vessels hydro-tested for maximum quality assurance.
- Provided with two locks one automatic and another is manual. The automatic lock operates when pressure gauge reads 0.30-0.37kg/Cm2d in the chamber.

Salient Features :

- Accurate in sterilization time and temperature.
- Door interlocks in case of double door sterilizer.
- Over heat protection
- Excess pressure safety valve.
- Self Diagnosis fault message.
- Safety against electric overload.
- Visual and audio alarm for easy operation.
- Process online print for records.
- Assured sterilization followed by complete drying of load.

Sources of Steam Generation:

Following combination of heating models are available.

- Electrically operated
- Steam operated (Optional)
- Combination of electric & steam operation.

Steam Generator:

A stainless steel electric boiler is provided under the shell with interconnecting steam pipes. The boiler is supplied with 6 KW immersion heaters mounted on stainless steel plate and electric controls would comprise of an air break contactor, automatic pressure controller, low water protection. The boiler also additionally provided with one independent spring - loaded safety valve and a pressure gauge. The boiler is also incorporates a gauge glass assembly with protector, water drain and water inlet valves are also provided.

Other Features :

- Two Nos. Mechanical ejectors
- Self Sterilizing vacuum drier
- Safety valve spring loaded and vacuum breaker
- Screen plug and Dial Thermometer for chamber's discharge line which is provided with steam trap and swing check valve
- Manual single point operation to operate manually as per IS 3829 Part I (Optional).
- Door safety - when the chamber is under pressure self locking device automatically locks and unlocks when chamber pressure falls below 0.35 k.g./cm2.

- Built in steam generator made of S.S. 304 with Low water cut out water level indicator self locking gauge glass valve.
- The equipment is provided with S.S. 304 railing and carriage with MS Trolley (Optional).
- Provided with vacuum system, low water cut off system.
- Suitable to run on inbuilt electric Boiler as well as from centralized steam supply available near to equipment @ 2.1 to 2.5 kg/cm2
- Working temperature 134°C.

Utilities & Environment Conditions :

- Supply Voltage 415 V, 3 Phase 50Hz power.
- Feed water : Soft water /DM Water/Good Water.
- Temperature range of +5°C to +45°C
- Maximum relative humidity of 80% for temperatures upto 31°C, decreasing linearly to 50% at 40°C.
- Main supply voltage fluctuation of + 5% of nominal.
- Provided with all piping connectors of SS:304

Automatic Process Control :

- Microprocessor based programmable logic controller is used to control the Process. Manual operation is also provided to avoid inconvenience in case of automation failure.

Programs Provided :

- Liquid cycle at 121°C
- Instrument Cycle 134°C
- Pre-Vacuum pulsating cycle
- Vacuum Leak Test.
- Bowie Test
- Flexible cycle from 105 to 134°C to suit users requirement.

Sizes available :

Dia	x	Depth	Load
400	x	600 mm	6 kw
400	x	1100 mm	9kw
500	x	900 mm	9kw
500	x	1200 mm	9kw

- Other sizes as per customer requirement

Other Optional Accessories :

- Sterilization Tray. Sterilization Basket.
- Carriage. Transfer Trolley.
- Air Compressor Audiovisual alarm for low water level.
- Autofeed water system. Temperature recorder.
- Thermograph or Printer for documentation
- Others as per requirement of the customer.

MODEL NO. USPL-USS-206

USPL ETHYLENE OXIDE GAS STERILIZER

Introduction :

Sterilization by ETO involves alkylation of Micro organisms i.e. alternation of genetic cell material

Process variables that play an important role in the process are

- ETO Concentration
- Relative Humidity
- Temperature
- Exposure time

These factors are carefully monitored in order to ensure complete sterilization.

Applications of Gas Sterilization :

- In the health product and medical fields :
- Fiberoptic endoscope, surgical telescopes and other optical medical equipment.
- Ophthalmic instrument.
- Dental instruments.
- Plastic instruments (e.g. specula's, syringes,)
- Oxygen tenets anesthesia masks and circuits.
- Artificial kidney machines, heart lung machines, cardiac pacemakers.
- Rubber and plastic tubing (e.g. catheters), surgical gloves, sheets, tubing.
- Cotton balls and pads, swabs and bandages.
- Electrical equipment (e.g. drills, pumps, motors).
- Respirator and inhalation therapy supplies.
- Surgical staplers /staples, sutures, sharps.

Salient Features :

- Cubical cabinet type model easy to install, less space requirement, elegant in look.
- The chamber is sq. / rectangular in shape constructed of heavy duty stainless steel 304 / 316.
- Specially designed coils to maintained uniform temperature in the chamber.
- Compact hot water circulation system.
- Humidification arrangement.

Design:

Chamber

Chamber is made of stainless steel. The rectangular section of the chamber is re-inforced with U-Profile jacket welded around the chamber. The chamber is insulated by resin-bonded glasswool covered by M.S./S.S. and performance as per BS & IS standards.

- Flame — proof design of the sterilization chamber with no spark— generation devices being mounted on the machine.



- Specially designed gas scrubbers for converting Ethylene Oxide to environmentally friendly Ethylene Glycol.
- Emergency evacuation system for rapid evacuation of the gas in case of emergency.
- Autoclave working pressure of 30" Hg to 1.2 kg/cm².
- Autoclave is designed specially for cartage type 100% EO gas with necessary safety features.
- Door gasket design is expansion type silicon.

Doors :

The unit is provided with One /Two Swing/Sliding (Optional) doors moved horizontally

Doors are provided with the following features :

- Double gasket sealing for safer operations(Optional)
- Process lock to prevent opening of the door during the process.
- Door interlocking for double door Model to prevent simultaneous opening of both doors (Optional)
- Heating jackets to ensure that the internal surface of the door have the same temperature as the chamber

Hot Water Generator

The Generator is fabricated form S.S. 304 Sheets with industrial immersion heaters. The steam generator is provided with automatic pressure control and other safety features like low water cut off to safe—guard heaters, safety valve gauge glass.

Vacuum System

Rotary vacuum pump driven by 3 phase electric motor. The other salient features are :

- Settable evacuation rate to suit packing material
- Emergency evacuation phase for fast evacuation of gas incase of emergencies

Safety:

Following safety features have been provided to ensure maximum operator safety

- Door closing and sealing as a precondition for start of the process
- Process lock to prevent opening of the door when the process is ON
- Double gasket for improved & fail proof door-sealing
- HI / LO temperature and pressure alarms

Process Control System :

- All sterilization processes are controlled by a Microprocessor based Process control System (PCS)
- Due to the wide variety of products sterilized in our system, many of which have entirely different packaging materials or pack sizes etc. the process control must be capable of extreme flexibility. It is of great value to our machine users to be able to change and develop process programs to satisfy their needs for superior products quality at optimum running cost.
- All control system are provided with a user friendly Man Machine interface for communication with the Microprocessor. This microprocessor combines flexibility with accuracy. Some of the salient features which can be incorporated in the system are:

- Capability of storing and running upto 6 different programmes
- All process parameters can be easily changed
- Pass-word protection to prevent unauthorized access.
- RS-232 output for down—loading critical process parameters for report generation
- Data Acquisition Systems for process Log and report
- A Comprehensive selection of alarm functions are also available
- Medial failures (Gas, Water, Air, Steam, electricity)
- Temperature and pressure alarms,

Time too long for different phases.

Gas leakage in the work area,

Insufficient feeding of ETO Gas etc.

Documentation and Validation

On special request design and manufacture of sterilization plants can be done in accordance with EN-550 for CE Mark. Machines can be supplied with :

- Design Qualification Report (DQR)
- Installation Qualification Report (IQR)
- Factory Acceptance Test (FAT)
- Performance Qualification Report (PQR)
- All assistance for on site validation and audit also guaranteed.

Technical Data :

1. Main Feature

1.1 Item	E.O. Gas Sterilizer
1.2 Chamber Size	410 x410 x 850 (5cft.) 450 x450 x1200 (8.5 cft.) Other sizes as per customer requirement
1.3 Sterilization medium	100%/EO Gas (Single use disposable cartage)
1.4 Material	Chamber : SS 304 /316 Door : SS 304 /316 Piping : SS 304 /316 Gasket : Silicon Paneling : MS Powder coated / SS 304
1.5 Controls	Microprocessor base control LCD Display : Chamber Heating, Pressure, Vacuum, Humidification, Gas Injection, Sterilization, Aeration & Sterile
1.6 Program	The following programs are located in the PLC with automatic aeration cycle 1. Warm Cycle at 55 to 60° C for 220 to 250 Minutes 2. Cool Cycle at 35 to 45° C for 300 to 350 Minutes 3. Selectable Cycle with programmable parameters
1.7 Accessories	1. Aeration hood built in 2. Sealing machine with timer up to 350mm sealing width (optional) 3. Air compressor (optional) 4. Gas exhaust treatment/ dosing unit built in 5. Electronic gas leakage detection system (optional)
1.8 Recording of Operational parameters	Built in printer (optional) for recording cycle parameter such as Temperature, Vacuum, Batch No., Aeration time etc.
1.9 Utilities & Environment	Power supply : 230 V Single Phase 50 Hz. AC Distill Water : 2 Lts. / Cycle Overall Sizes : 850 x 1600 x 1120 / 1000 x 1700 x1500 (Wx H x D) Air Pressure : 5 kg. /cm' max. 50 lpm



MODEL NO. USPL-USS-208

USPL AUTOMATIC H.P.H.V. RECTANGULAR HORIZONTAL STERILIZER

Construction :

- The Chamber is sq./rectangular in shape constructed of heavy duty stainless steel 304/316 door and jacket of stainless steel 304 (316L or 316Ti-optional)
- The Pipes and fittings of stainless steel and bronze, the valves are ball type. Each sterilizer is mounted on a M.S/S.S sturdy stand. The outside paneling is made of M.S Powder coated or S.S to give elegant look. Hinged panels on sider makes it friendly for service purpose. All other construction details are as per IS 3892 Part 1 199.

Design :

- Steri autoclave (YSU-608) is designed as per international standard.
- Chamber of the autoclave is tested for hydraulic test pressure, performance as per BS & IS standards.
- Working pressure of autoclave is from 1.2 to 2.1 kg. cm² and vacuum upto 26"HG (with built in vacuum pump arrangement).
- Door gasket is expansion type silicon and jointless.
- The heating system provided is energy efficient and

reduces running cost to great extent.

- Sterilization process is controlled by programmable logic controller with high accuracy & takes care of all user requirements.

Pulsating Deep Vacuum :

- High efficiency vacuum pump for mechanical air removal for excellent steam penetration even in fabric loads.
- Post sterilization vacuum drying treatment to ensure the load is dry on unloading.

The chamber is designed and tested as per American Society of Mechanical Engineers standards.

Doors

- Available Single and double door model offers the customer two door options - hinged door and automatic sliding door (Optional). All doors have stainless steel contact parts and temperature resistant silicon rubber gasket.

Recording System (Optional)

- Single point strip chart recorder for recording the process temperature, date and time.

Safety :

- Door safety to prevent starting of the process unless the door is closed and opening of the door when the chamber is pressurized (optional).
- Insulated chamber to avoid scalding to the operator.
- Safety valves for over-pressure safety.
- Pressure vessels hydro-tested for maximum quality assurance.
- Provided with two locks one automatic and another manual. The automatic lock operates when pressure gauge reads 0.30-0.37 kg/cm² in the chamber

Salient Features :

- Accurate in sterilization time and temperature.
- Door interlocks in case of double door sterilizer.
- Over heat protection
- Excess pressure safety valve.
- Self Diagnosis fault message.
- Safety against electric overload.
- Visual and audio alarm for easy operation.
- Process online print for records.
- Assured sterilization followed by complete drying of load.

Sources of Steam Generation :

Following combination of heating models are available.

- Electricity operated
- Steam operated (Optional)
- Combination of electric & steam operation.

Steam Generator:

A stainless steel electric boiler is provided under the shell with interconnecting steam pipes. The boiler is supplied with 6 KW immersion heaters mounted on stainless steel plate and electric controls would comprise of an air break contractor, automatic pressure controller, low water protection. The boiler is also additionally provided with one independent spring - loaded safety valve and a pressure gauge. The boiler also incorporates a gauge glass assembly with protector, water drain and water inlet valves are also provided.

Other Features :

- Two Nos. mechanical ejectors
- Self Sterilizing vacuum drier
- Safety valve spring loaded and vacuum breaker
- Screen plug and Dial Thermometer for chamber's discharge line which is provided with steam trap and swing check valve
- Manual single point operation to operate manually as per IS 3829 Part I (Optional).
- Door safety - when the chamber is under pressure self locking device automatically locks and unlocks when chamber pressure falls below 0.35 k.g./cm².
- Built in steam generator made of S.S. 304 with Low water cut out water level indicator self locking gauge glass valve.

- The equipment is provided with S.S. 304 railing and carriage with MS Trolley (Optional).
- Provided with settable, vacuum system, low water cut off system.
- Suitable to run on inbuilt electric Boiler as well as from centralised steam supply available near to equipment @ 2.1 to 2.5 kg/cm²
- Working temperature 134° C.

Automatic Process Control :

- Microprocessor based programmable logic controller is used to control the Process. Manual operation is also provided to avoid inconvenience in case of automation failure (Optional).

Programs Provided :

- Liquid cycle at 121°C
 - Instrument Cycle 134°C
 - Pre-Vacuum pulsating cycle
 - Vacuum Leak Test.
 - Bowie Test
 - Flexible cycle from 105 to 134°C to suit user requirement.
- Utilities & Environment Conditions
- Supply Voltage 415 V, 3 Phase, 50Hz, for power refer chart
 - Feed water : Soft water/DM Water/Good Water.
 - Temperature range of +5°C to +45°C
 - Maximum relative humidity of 80% for temperatures upto 31°C, decreasing linearly to 50% at 40°C.
 - Main supply voltage fluctuation of + 5% of nominal.
 - Provided with all piping connectors of SS:304

Sizes available:

Width (mm)	Length (mm)	Height (mm)	Electric Load (K.W.)
450	450	900	18
600	600	1200	18
600	900	1200	36
910	910	1200	36
600	900	1500	36

Other sizes as per customer requirement.

Other Optional Accessories :

- Sterilization Tray.
- Sterilization Basket.
- Carriage.
- Transfer Trolley.
- Air Compressor
- Audio visual alarm for low water level.
- Auto feed water system.
- Temperature recorder.
- Thermograph or Printer for documentation Others as per requirement of the customer.



MODEL NO.: USPL-USS-608ex

USPL AUTOMATIC SLIDING DOOR HPHV RECTANGULAR H. STERILIZER

Features :

- YSI Automatic sliding door steam sterilizer
- This sterilizer is an advanced version of YSU 608 and is the ideal solution for hospitals / Pharmaceuticals / Research Institutes sterilization needs.

Engineered with people in mind :

Features enable the convenience and durability needed to operate a sterilizer with complete peace of mind. The jacketed chamber is constructed of long lasting 304/316L grade stainless steel with superior corrosion resistance. In addition, the generator and piping are constructed of stainless steel. The pneumatic valves are air pressure operated, significantly reducing maintenance. They are safe and reliable, eliminating the requirement for high voltage. To save energy, the sterilizers automatically switch to a standby mode.

Control Made Simple :

Uses an advanced microprocessor for the automated process control. Control panel is user-friendly consists of LCD and a functional touch pad. A self diagnostic system provides messages and alerts that are displayed in a clear language. Audible alerts are sounded with the messages. The LCD display provides constant cycle parameter readouts and cycle progress information. The control system is self diagnostics and provides fault messages to the operator.

The Control system is programmed with a total of 7 programs or more (Optional) that range in temperature from 105°C to 134°C

- Programs for (unwrapped instruments)
- Programs for wrapped instruments and packs
- Slow exhaust programs.
- Test programs Bowie & Dick and vacuum Leak Test
- If needed the number of programs can be increased to take care of user requirement.



Safe Design :

Door Safety

The doors are designed with independent mechanical and electronic safety features that guarantee a safe working of the sterilizer.

- A safety device prevents the operator from opening the door when chamber is pressurized.
- Steam is not allowed into the chamber when the door is open.
- A cycle will not start if the door is open or not properly closed.
- The door cannot open until chamber pressure reaches room pressure.
- Sliding Door Safety-The sliding door will automatically stops if an obstruction is detected.
- Double Door Safety- interlocks prevent both doors from being opened simultaneously.

General/ Safety Features

- Double Independent Monitoring : The combined digital and mechanical monitoring provides a cross reference and guarantees accurate results. The operator has two independent means to monitor temperature and pressure.
- Safety Valves : Both the chamber and the jacket are equipped with safety valves-if the pressure exceeds the allowed limit the safety valves will discharge.
- Built-in Steam Generator Safety : A water level monitoring system maintains a constant water level and ensures safe operation of the heaters.
- Emergency Shut-off : Easily accessible emergency switches for immediate and safe shut down of the sterilizer.

1. Main Feature:

1.1 Item	HIGH PRESSURE HIGH VACUUM STEAM STERILIZER FULLY AUTOMATIC, SLIDING DOOR SINGLE/DOUBLE
1.2 Purpose	For treatment of Sterilization of O.T. Material
1.3 Chamber Size	600 (W) x 600 (H) x 1200 (D) mm to 1200 (W) x 1500 (H)x 2400 (D) mm
1.4 Chamber Volume	460 to 4300 Ltrs.
1.5 Working Temp.	Upto 134°C
1.6 Working Pr.	1 to 2.2 k.g./cm ²
1.7 Chamber test Pr.	3.5 k.g./cm ²
1.8 Jacket test Pr.	4.8 k.g./cm ² , Channel Jacket.
1.9 Source of Energy	Electrically Operated/Direct Steam Operated
1.10 Mounting	Floor Standing Horizontal Type

2. Design

2.1 YSU-608Ex. 205	Steri Steam Sterilizer are designed to meet the norms laid down by the "ASME, BIS, IS 302, EN issued and equivalent international standards for Steam Sterilization dedicated Autoclaves.
2.2 Level of Sterilization	Less than 10 ⁻⁶ , over kill approach which results in total destruction of bacillus stearothermophilus.
2.3 Standards	Shell Fabricated & tested in accordance with IS 3829 Part I & ASME Sec VIII Division I & as per other International standards.
2.4 Certification	Quality Assurance (QA) certification which compliance with requirements of ISO 9001-2000 by BSI affiliated to Ukas & ANSI-RAB, CE Marked.

3. Construction Features

3.1 Cross Section	Square/ Rectangular
3.2 Shape	Rectangular
3.3 Welding	Butt Fusion welded (Argon Arc)
3.4 Material	Chamber : AISI 304/316/316L Channel Jacket : Boiler Grader Steel/AISI 304/316/316L Outer Cover/ Panelling : AISI 304 /Aluminum Baffle : AISI 304 Piping : AISI 304 Door : AISI 304/316/316L

4. Door

4.1 Type	Sliding Single or Double Door
4.2 Sealing	Silicon Tubular Gasket (durable enough to with stand inside temperature & pressure as well as hydraulic test pressure)

5. Steam Generator

Electrically Operated

5.1 Source of Energy

5.2 Capacity

Adequate/Sufficient for complete sterilization cycles

5.3 Fittings

a) Water Level Indicator/ Sight Glass

b) Water inlet & outlet valve

5.4 Safety

a) Pressure Regulating Electric Device

b) Spring loaded safety valve

c) Low water protection for heaters to cut of the supply if water level falls below the minimum level.

d) Self locking gauge glass valve to protect electric circuit and human against thermal injuries in case of breakage.

6. Piping & Fittings

6.1 SS Piping	Made of stainless steel and bronze, duly argon arc welded.
6.2 Dial Thermometer	Fitted in the chamber drain line 6/10 cm diameter with $\pm 1.5^\circ$ accuracy.
6.3 Pressure Gauge	6.5 / 7.5 cm diameter industrial type pressure gauge mounted on the jacket.
6.4 Compound Gauge	6.5 / 7.5 cm diameter industrial type Compound gauge mounted on the Chamber.
6.5 Manual Operational Valve	Single Port operational valve in case of manual operation (optional).
6.6 Safety	a) Relief valve for jacket b) Vacuum breaker in the jacket to prevent vacuumization of jacket c) Bacteria retentive filter to avoid contamination of load fitted in the drying system

7. Vacuum Pump for Pre Post Vacuum Pulse

7.1 Type

Water-ring motorised vacuum pump, suitable for three phase electric supply. 380/50 Hz.

7.2 Purpose

Pre evacuation of chamber before sterilization as per norms and for drying sterilized load.

8. Accessories

8.1 Air Compressor (Optional) Provided of suitable capacity for all pneumatic operations.

8.2 Secondary Sterilization Device Provided with the autoclave

9. Controls

- | | | |
|-----|-----------------------|--|
| 9.1 | Panel | Steel/Stainless Steel cubical modular type panel incorporating various pressure gauges to indicate actual pressure in chamber jacket and on gasket. |
| 9.2 | Control Accessories | Circuit breaker, Overload relay, Transmitter, Switches, Indicator, Push Buttons & Connectors mounted inside the control panel |
| 9.3 | Process control | PLC with four line LCD or more with numeric and command key to feed alphanumeric data. |
| 9.4 | Man Machine Interface | Userfriendly Alpha-numerical / Graphical / Digital type display |
| 9.5 | Display | Cycle status Fault/Error Indication with visual alarms |
| 9.6 | Program | The following programs are loaded in the PLC with one pre vacuum pulse to purge the autoclave of all air under strict Sterilization Control cycle software program.
a) Temperature of not less than 121°C and a pressure of 15 psi for an autoclave residence time of not less than 20 minutes for each cycle
b) Temperature of not less than 135°C and a pressure of 31 psi for an autoclave residence time of not less than 7 minutes for each cycle.
c) Validation test program for periodic validation. |



10. Recording of Operational Parameters (Optional)

- | | | |
|------|---|--|
| 10.1 | Computer Recording
Microprocessor controlled | Recording of process value through a dot matrix printer on a roller paper as per real time printing device to record events during continuation of cycle. The event recording include exact hour, minute and seconds of following parameters
a) Chamber Temperature b) Chamber Pressure
c) Selected Program d) Cycle Status or Cycle Events
e) Date, Day, Year, Time f) Batch Identification facility
g) Fault & emergency message with code |
|------|---|--|

11. Test

- 11.1 Each equipment is tested for following tests before dispatch and certified
- a) Chemical test - A chemical indicator sterilization test as per biomedical rule-98.
 - b) Biological test - Bacillus stearothermophilus spore strips, with at least 1 x 10⁶ spores/m
 - c) Validation test - Periodic validation for recording the efficiency of the autoclave

12. Utilities & Environment Requirement

- | | | | | |
|------|-------------------|---|------------------------------------|--|
| 12.1 | Supply Voltage | : | 415 + 5% Volt 3 Phase | Inner Sizes |
| 12.2 | Frequency | : | 50 Hz. | 450 x 450 x 900mm |
| 12.3 | Power Consumption | : | 36 KW to 72 KW | 600 x 600 x 1200mm |
| 12.4 | Water Supply | : | Approx 150 ltrs to 300 Ltrs./cycle | 600 x 900 x 1500mm |
| 12.5 | Environment | : | 5 to 50°C | 900 x 900 x 1500mm |
| 12.6 | R.H. | : | 90% at 30°C | or any size as per customer requirement. |

- | | | | |
|-----|-------------------------------|---|--|
| 13. | Carriage & Trolley (Optional) | : | <p>Loading Equipment Our loading equipment assists for the loading and unloading process. It is made of high quality, durable stainless steel. We offer two options. Pull Out Trays : Stainless Steel trays equipped with rails for easy loading and unloading. The rails are designed to prevent the trays from rolling over.</p> <p>Loading Carts and Transfer Carriages : The adjustable loading cart rolls from the transfer carriage onto the interior chamber tracks for easy handling of heavy loads. The trolley is equipped with revolving wheels, maximizing mobility in limited space. The wheel breaks prevent the trolleys from rolling and the carriage is equipped with a lock that prevents it from sliding.</p> |
|-----|-------------------------------|---|--|

MODEL NO.: USPL-USS-309

USPL DISINFECTOMAT WASHER DISINFECTOR

General:

Hospital infection control is a serious problem which no Hospital Administrator can afford to ignore. Proper attention needs to be focused on the system of Cleaning, Disinfection, Packaging, Sterilization and Storage of different surgical appliances and instruments.

The continuous advancements made in the field of Biomedical Engineering have resulted in development of a range of surgical appliances and apparatus with very intricate and delicate profiles. Any sterile supply chain must have a system which can clean and disinfect these components with a high degree of accuracy and repeatability.

Steri "DISINFECTOMAT" Washer Disinfectors have been designed with just that in mind. The system is not only flexible, but is also very user friendly to take care of your every needs.

Salient Features :

- Ergonomic design.
- Suitable for efficient treatment of Large quantities of medical items in a centralized sterile supply department Convenient front loading.
- User friendly controls.

High Impact Cleaning System

The Disinfectomat is provided with three rotary spray arms. The design and location of the arms ensure that every surface of the load is uniformly cleaned. The unit is constructed with out any angles in order to prevent dirty breeding grounds. The rotation of the arm is activated by the water pressure which eliminates any mechanical drive system

- Inbuilt Disinfection System

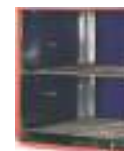
The unit is incorporated with an in-built electric water heater. The hot water here is used to disinfect the load at a temperature of 90°C (i.e. hemodisinfection) after thorough cleaning and rinsing.

- Flexible Loading System

Steri Disinfectomat comes with a range of standard accessories for different surgical appliance & Kidney Trays, Bottles, Tubings and instruments. Additional loading cassettes are available as optional

- Automatic Detergent Dozing

The machine is incorporated with an automatic detergent



dozing system to doze the required amount of detergent in the rinse water.

User Friendly Program Menu

The Disinfectomat has a Microprocessor Controlled operating system. The operating Console is provided with a membrane key pad and electronic display unit of easy operation and monitoring of the process. Up to 3 pre-programmed processes give the user fantastic flexibility for cleaning and disinfecting a wide range of load configurations

In-Built Heating System

In Hospitals where hot water is not available the Disinfectomat is equipped with an electric heating system. This heats the water to a temperature of 90°C for higher cleaning efficiency.

Fully Sanitary Construction

The body is fabricated from non-corrosive 304 Grade Stainless Steel with high quality argon welding. All corners are rounded for easy cleaning and to avoid contamination. The door is provided with a food grade silicone gasket. All piping and valves are of stainless steel construction with silicon seals.

Installation

The Disinfectomat is available in both single door and double door options to suit different CSSD configurations and requirements.

The machine requires no installation. It has to be connected to the different supply lines and drain to start operation

MODEL NO : USPL-USS-309ex

USPL DISINFECTOMAT WASHER DISINFECTOR

SALIENT FEATURES : -

- Cycle pump with large flux can wash utensils efficiency
- The cycle water can be heated to 90oC and disinfect goods primarily
- 6 washing program automatically
- High efficiency dry performance
- Auto adding detergent
- Intelligent control system
- Auto heating function
- Auto water discharging
- Safety door device with chain control
- Temperature insulate
- Washing chart with large volume (Optional)

Designed with large storage surface. An excellent complement to the existing installation. Takes over the connection of existing old appliances and can be extended by means of accessories such as Bed Pan and Urine Bottle shelves to complete the hospital cleaning center. Most suitable for emptying, cleaning and disinfecting care utensils such as Bed Pan urine bottle, commode buckets, receptacles etc. All Technical parameters are same as YSU-609 but with Double.

• High Impact Cleaning System:

The Disinfectomat is provided with three rotary spray arms. The design and location of the arms ensure that every surface of the load is uniformly cleaned. The unit is constructed with out any angles in order to prevent dirty breeding grounds. The rotation of the arm is activated by the water pressure which eliminates any mechanical drive system

• Inbuilt Disinfection System:

The unit is incorporated with an in-built electric water heater. The hot water here is used to disinfect the load at a temperature of 90°C (i.e. hemodisinfection) after thorough cleaning and rinsing.

• Flexible Loading System:

Steri Disinfectomat comes with a range of standard accessories for different surgical appliance & Kidney Trays, Bottles, Tubings and instruments. Additional loading cassettes are available as optional

• Automatic Detergent Dozing:

The machine is incorporated with an automatic detergent



dozing system to doze the required amount of detergent in the rinse water

• User Friendly Program Menu:

The Disinfectomat has a Microprocessor Controlled operating system. The operating Console is provided with a membrane key pad and electronic display unit of easy operation and monitoring of the process. Up to 3 pre-programmed processes give the user fantastic flexibility for cleaning and disinfecting a wide range of load configurations

• In-Built Heating System:

In Hospitals where hot water is not available the Disinfectomat is equipped with an electric heating system. This heats the water to a temperature of 90°C for higher cleaning efficiency.

• Fully Sanitary Construction:

The body is fabricated from non-corrosive 304 Grade Stainless Steel with high quality argon welding. All corners are rounded for easy cleaning and to avoid contamination. The door is provided with a food grade silicone gasket. All piping and valves are of stainless steel construction with silicon seals.

• Installation:

The Disinfectomat is available in both single door and double door options to suit different CSSD configurations and requirements. The machine requires no installation. It has to be connected to the different supply lines and drain to start operation

MODEL NO : USPL-USS-317

USPL TABLE TOP PORTABLE AUTOCLAVE

Elegantly designed for use in Hospitals, Clinics, Dispensaries, Pharmaceuticals, and R & D Department to Sterilize the surgical instruments/items during operation, media, culture etc or to inactivate the bacteria from reproduction at various stages.

This Sterilizer is specially designed for causality O.T. & for dental department by seeing the present demand. Fitted with automatic devices for time and temperature selection. Various control for Boiler Temperature/Pressure Controller are also provided alongwith manual chamber relief valve in case of power failure, buzzer sound to indicate sterilization cycle is over.

Construction : S.S. 304 Chamber, Steam Generator & Jacket alongwith Chamber Lid. Outer body made of M.S. duly powder coated or SS. Fitted with 1/2" BSP male connection for drain and 1/2" BSP mail connection for exhaust



Suitable to work on 230 Volts, Single Phase, 3 K.W. power supply 50 Hz, AC supply.

Available in various chamber sizes.

Chamber capacity - 14 Lt.

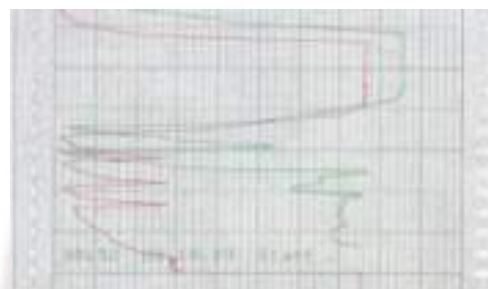
Size : 210 Dia x400 mm D

Power : 230 VAC 50 Hz, 2 KW

Chamber capacity - 19 Lt.

Size : 210 Dia x550 mm D

Power : 230 VAC 50 Hz, 4 KW



GAUZE CUTTING MACHINE

MODEL NO : USPL-USS-G-100

- Useful in cutting thickest of cotton gauze material.
- Consist of a cutting unit and a knife sharpening unit.
- Blade size approx. 200 mm.
- Cutting Capacity 165 mm.
- Work on 230 V, 50 Hz power supply. 15 no's blades for gauze cutting machine.



MODEL NO : USPL-USS-310

USPL SINGLE WATER STILL

As per IS -3830

Free standing, electrically operated water still that is capable of producing Pyrogen free distilled water as per IP/BP standards. All contact parts are made of stainless steel. fitted with ISI Marked Immersion water heater, low water protection & electrical control box. It is mounted on a sturdy MS tubular stand.

Capacity: 5 Lit/ hr, 10 Lit/hr, 20 Lit/hr, 40 Lit/hr, 60 Lit/hr.

Automatic version also available.



MODEL NO : USPL-USS-310-2

USPL DOUBLE WATER STILL

As per IS -3830

Free standing, electrically operated double water still capable of producing Pyrogen free distilled water as per IP/BP standards. All contact parts are made of stainless steel. This comes with ISI Marked Immersion water heater, low water protection & electrical control box. It is mounted on a sturdy MS tubular stand.

Capacity: 5 Lit/ hr, 10 Lit/hr, 20 Lit/hr, 40 Lit/hr. Automatic version also available.



WASH STATIONS WITH 2 SINKS

MODEL NO : USPL-USS-WS-001

Backed by rich industry knowledge, we are offering our customers an excellent quality of BMW Shredders. In order to design these shredders, The offered shredders are broadly applauded by our customers for their quality. Our customers can take these shredders from us at market-leading rates.



FEATURES

- ✚ Size Approx. (LxWxH): 200x900x700 mm with sink sizes of 40x500x250mm(wdh)
- ✚ Water shower/water spray gun attachable.
- ✚ Air Gun attachable
- ✚ Made of solid, bright-polished stainless steel (304) to withstand heavy-duty work with the wet instrument. Designed with a 60 mm high edge (splash back) at the rear. The front and side edges are reinforced and widened to 49 mm. Edges are welded together and polished at the corners.
- ✚ The worktop slope to the sink and reinforced by a full-length support frame.
- ✚ Sink units of sizes that allow processing of the large modular instrument trays.
- ✚ The legs are able to provide strong support and hold to the entire unit securely.
- ✚ The Sink includes a drain valve, removable strainer, manually operated drain valve, overflow drainpipe, and water trap. The table includes a mixing faucet with a swivel spout for cold and hot water connection.

WORK TABLE

MODEL NO : USPL-USS-WT-100

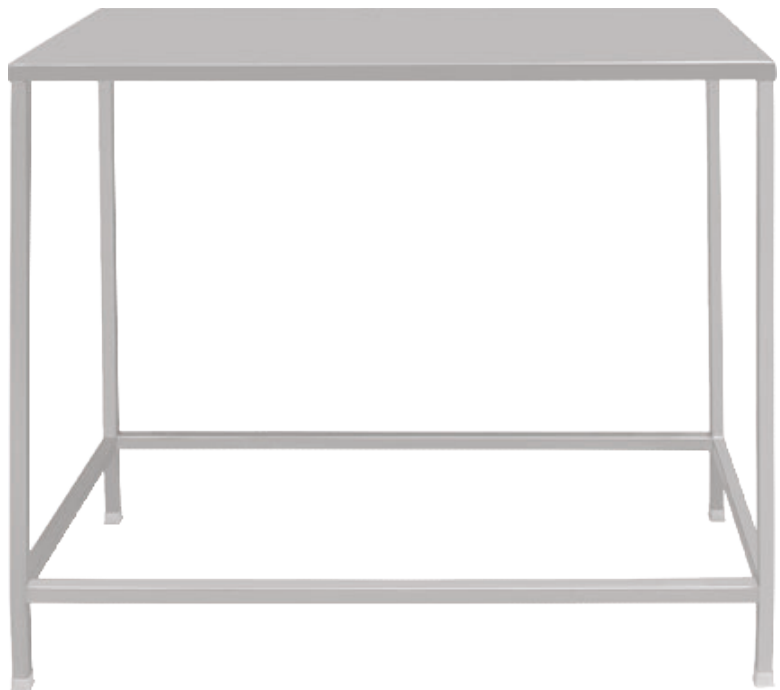
FEATURES

- ☞ Size Approx. (LxWxH): 1200x650x900 mm approximately.
- ☞ Stainless steel table especially designed for inspection and sorting of wet goods in heavy-duty areas and for general purpose pre-storage.
- ☞ The work tables a rigid stainless-steel construction which is easy to clean and does not have sharp edges or corners.
- ☞ The table ergonomically worked up, have easy to clean robust matt-finished (to reduce reaction of light from the surface) with minimum sheet thickness of 1.55 mm stainless steel (304) worktop /surface to withstand and carry out heavy work comfortably, either sitting or standing.
- ☞ They are welded together and polished at all corners for good hygiene, as well as for the comfort and safety of the staff.
- ☞ The worktop is supported by a complete assembly with full-length reinforcements along the front, back, and welded together at the corners.
- ☞ All edges will be smooth, and the rigid frame made up of a minimum 1.5 mm sheet thickness stainless (304).
- ☞ Unobstructed access to the working space, since the only supports needed along the front of the table are the corner legs. This also facilitates the cleaning of floors.



CONTROL & PACKING TABLE

MODEL NO : USPL-USS-WT-100



FEATURES

- ✎ Size (L x W x H) : 2000x1400x1400 mm approximately.
- ✎ This table is specially designed for sorting, inspection, functional control, and packing of various sets for wards, clinics , etc., and for surgical instrument sets in trays. The work could be done comfortably, either sitting or standing.
- ✎ The worktop is made of robust wood-based core material surfaced with plastic laminate in a soft beige color that reduces reflection of light from the surface. All edges are smooth. The extended w of the worktop is designed to facilitate thorough inspection of instrument trays and allow the use of large wrapping material.
- ✎ The rigid frame is made of stainless steel (304)
- ✎ There is unobstructed access to the working space since the only supports needed along with the front of the table are the corner legs. This also facilitates the cleaning of floors.
- ✎ Have double workspace. One workplace table has a 700 mm wide worktop, and the other workplace has 1400 mm worktop.
- ✎ The table includes a two-shelf console, mounted on the worktop, for storage of packaging materials.
- ✎ The rigid supporting columns of the console include 3 electrical outlets.
- ✎ There is a free space of 450 mm between the lower shelf and the worktop and 150 mm between the two shelves.
- ✎ The table has a drawer unit (both sides as a double model) mounted under the worktop.
- ✎ Each drawer unit 400 mm wide and include a drawer and a sliding plate.
- ✎ Fluorescent tube fittings (inspection lamp) available.

PASS BOX

MODEL NO : USPL-USS-PS-001

FEATURES

- ✚ Area: Dirty to Clean, Clean to Sterile & Sterile Issue.
- ✚ Size: 600x600x600 mm, internal
- ✚ Made up of SS 304 sheets with the double-wall construction.
- ✚ Door interlocking to prevent the simultaneous opening of both the doors.
- ✚ Toughened glass paneling for easy visibility.
- ✚ ISO 9001 and ISO 18001 certified.



DISTILLATION UNIT

MODEL NO : USPL-USS-DU-001

FEATURES

- ✚ Floor Mounting Type
- ✚ Widely used in Hospitals/ Laboratory and Pharmaceutical Industries.
- ✚ Single, double, and triple Distilling Units, operated by electricity/direct system.
- ✚ Fully made of SS 304 or 316 with storage tank.
- ✚ Capacity: 5 to 600 liters per hour



LINEN FOLD TABLE FOR CLEAN AREA

MODEL NO : USPL-USS-LFT-001



FEATURES

- ✚ Size 2000 x 1400 x 900 mm
- ✚ The table is specially designed for sorting, inspection (each piece of linen can be moved over an illuminated inspection panel) and folding of surgical dressing sets and individually packaged towels/gowns. The extended width also facilitates work with large dressing sheets. Work can be carried out comfortably, either sitting or standing.
- ✚ The worktop made of a robust wood-based core material surfaced with plastic laminate in a soft white color that enhances the lighting for inspection of linen.
- ✚ All edges of the worktop smooth.

CRUB STATION

MODEL NO : USPL-USS-SCS-001

Our firm is betrothed in on varied industry parameters. Presenting Wash Station with Two Sinks to our valuable client. The unit consists of two sinks. These products are manufactured employing the finest raw material that is obtained from proficient industry sellers. These products are extremely admired in the market due to their superior finish, low maintenance, and longer operational life. Top keep the quality; our provided products are thoroughly checked.

CONSTRUCTION SALIENT FEATURES

- ✚ Longer working life
- ✚ Easy to use
- ✚ Sturdy construction
- ✚ Stainless steel construct
- ✚ Pre-painted GI sheets
- ✚ Compact yet spacious design
- ✚ Easy maintenance
- ✚ Corrosion resistant
- ✚ Durable
- ✚ Dimensional stability



RACK & TROLLEY

MODEL NO : USPL-USS-R-001

All sizes and specifications of SS Racks are available with us and made up with the stainless steel materials. It is equipped by our team with the multiple shelves and foldable as well our racks can be adjusted at any places of clinics easily. Regarding quality, our offered SS Rack Manufacture is tough and non breakable in nature.





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